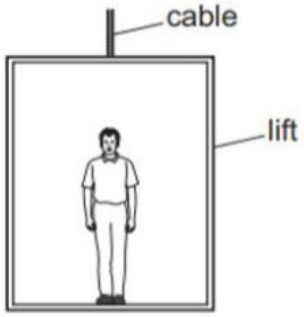
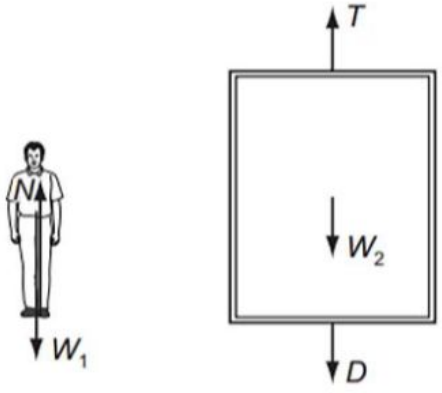
4 The diagram shows a man standing in a lift.



The forces acting on the man and the forces acting on the lift are shown below.



N is the force from the lift floor on the man.
W_1 is the weight of the man.
T is the tension in the lift cable.
W_2 is the weight of the lift.
D is the force from the man on the lift floor.

Which of the following statements is correct?

A $(W_1 + W_2)$ is always equal to T.

B If $N = W_1$, the lift must be at rest.

C N and W_1 are always equal and opposite.

D If $T = (D + W_2)$, the lift must have a constant velocity.

A: This would only be true if the lift system has zero acceleration. If the lift accelerates, T must be greater or less than $(W_1 + W_2)$. So, this is incorrect.

B: For the man: $N - W_1 = ma$. If $N = W_1$, then $a = 0$, meaning zero acceleration — but this includes constant velocity motion, not just rest. So, this is incorrect.

C: N and W_1 are only equal when acceleration is zero. When the lift accelerates, $N \neq W_1$. Also, they are not a Newton's 3rd law pair (they both act on the same object — the man). So, this is incorrect.

Ans: D

Applying Newton's 2nd law to the **lift alone**:

$$T - W_2 - D = m_{\text{lift}} \times a$$

If $T = D + W_2$, then the net force on the lift = 0, meaning $\boldsymbol{a = 0}$, so the lift moves at constant velocity (or is at rest). This is **correct**.